Week 4 Progress Report
BSE4104 Emerging Trends in Software Engineering
AI-Native & Agentic Engineering
Capstone Project
SME Procurement-Preparation Agent
GROUP J EVENING

1. **Work Completed Against Weekly Objectives**
    - Defined a clear Tool Catalogue containing three tools:
        - get_inventory (read-only)
        - get_quotations (read-only)
        - create_draft_requisition (low-risk side-effect)
    - Designed complete input and output JSON schemas, authorization rules and failure behaviour for each tool.
    - Implemented the actual tool functions in Python with proper validation and error handling.
    - Integrated the tools with the foundation model using OpenAI function calling.
    - Built a working orchestrator that allows the model to decide when to call tools, execute them, and return grounded results.
    - Added structured logging of every tool call (arguments, results, duration, success/failure) and saved traces to evidence for observability.
    - Designed and documented failure and authorization test cases covering validation errors, not-found cases, safety boundary violations and robustness scenarios.
    - Updated the system architecture to include the Tools layer and the Human Approval Gate.
    - Committed all code, schemas, synthetic data, and test documentation to the GitHub repository.

    The system can now move beyond pure text generation and perform useful, controlled actions through explicit tools while keeping high-impact decisions under human control.

2. **Key Engineering Decisions and Rationale**

Decision	Rationale
Limited to three focused tools	Keeps the system small and explainable while still meeting the “at least two tools” requirement.
create_draft_requisition only creates Draft status	Satisfies the safety boundary: the agent can prepare work but cannot approve or commit.
Strict input validation inside each tool	Prevents bad data from entering the system and produces clear, testable error messages.
Full structured logging of every tool call	Provides strong evidence for evaluation, debugging, and the Week 7 observability requirements.
No tool exists for “approve” or “place order”	Hard technical enforcement of the AI Boundary Matrix.

3. **Challenges Encountered and Current Response**

Challenge	Response
Ensuring the model does not invent data when tools are available	System prompt strongly instructs the model to always use tools for current data.
Handling invalid or missing parameters cleanly	All tools return structured error objects instead of crashing.
Demonstrating authorization boundaries	Created explicit test cases (TC-09 and TC-10) that attempt prohibited actions and confirm refusal.
Keeping traces useful but not overly verbose	Logged only essential fields (tool name, arguments, result, duration, success).

4. **Repository and Task Management Evidence**

GitHub Repository: https://github.com/Nicholas-Muwonge/SME-Procurement-Agent

Key additions this week:

- src/tools.py - Full implementation of the three tools + OpenAI tool definitions
- src/orchestrator_with_logging.py - Function-calling loop with trace logging
- data/inventory.json and data/quotations.json - Synthetic data
- docs/tools/Tool_Catalogue_and_Schemas.pdf
- docs/evaluation/Week4_Tool_Failure_Authorization_Tests.pdf

ClickUp Board: https://app.clickup.com/1200420000004171/v/li/1200420000007246

5. Plan for Week 5

Focus: *Agent Architecture and Bounded Autonomy*

- Define one clear multi-step goal that benefits from decision-making across tools.
- Design the Sense - Plan - Act - Observe - Stop loop.
- Set maximum iterations, approved tools, stop conditions and human hand-off rules.
- Implement the bounded agent workflow.
- Capture at least three execution traces including one failure/recovery case.
- Deliverables: Agent Architecture Diagram, Agent Task Contract, working bounded agent, three execution tracesand Week 5 Progress Report.